Amazon Product Analysis

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Abstract

Amazon is a multinational e-commerce company and has experienced a 41% decline in product purchases during its Prime Day event in 2025 compared to the previous year. As a result, both the *Marketing* and *Sales* departments at Amazon are looking to improve their customer experience. They seek the help of the Lead Data Analyst to provide them with a comprehensive product report which will examine the products carried by Amazon. The *Marketing* department will use this information to deliver product-specific digital promotion while the *Sales* department will use this information to suggest a range of products most suitable to each customer.

Research Questions

A product analysis will be performed for the marketing and sales departments with a focus on answering the following research questions.

Question 1
How does the price, rating and discount compare for products belonging to different categories?

Purpose
Obtain an overview of the products sold by Amazon to make it easier to implement guided marketing and sales strategies.

Impact Hypothesis
When a sales employee is making efforts to sell a product to a future customer, the employee will have sufficient knowledge to answer any questions about the product, including questions relating to pricing and potential discounts.

Question 2
How can the product characteristics be used to predict the product category?

Purpose
Classify new products to ensure appropriate marketing and sales strategies are applied for market success.

Impact Hypothesis
A member of the *Marketing* team is asked to curate digital content to promote a new product. In this case product category classifier models can be used to identify the already existing digital materials prepared for products belonging to the same category.

Question 3
How can product descriptions be used to identify similar products?

Purpose

Match new products to existing products for segmentation and recommendation strategies.

Impact Hypothesis
A sales employee needs to find products similar to the one described by a customer using the product information retrieval models.

Literature Review

Predicting Purchase Intention Categories Using Logistic Regression

E-commerce operations expand beyond the online store itself, with advertisements for products spreading across various social media platforms. While data from such platforms can provide key insights on user behaviour, it can be difficult to accurately analyze the hidden effects of the inherent time-series data (Wu, 2025). Wu proposes the use of machine learning algorithms and their abilities to capture complex patterns in the data to predict the category of users' purchase intentions. These categories include those which are common to many large e-commerce retailers today such as beauty, digital, food, clothing, etc. While structured user data can be used to build and train models like logistic regression, the lack of temporal and textual data can hinder prediction accuracy and marketing strategies (Wu, 2026).

Wu proposes the use of a Transformer-BiGRU classification algorithm for its ability to handle temporal dependencies in behavioural data unlike traditional models. The study found that the proposed predictive model outperformed all other models when comparing evaluation metrics. Indeed, as suggested prior by Wu, the logistic regression model delivered nearly the lowest performance for accuracy, precision and recall. Notably, while the decision tree and KNN models were not as effective as the Transformer-BiGRU, they still recorded higher metrics than the logistic regression model.

This study validates the use of a multinomial logistic regression model in the case of predicting retail categories in an e-commerce setting. Although this study is focused on using user data to predict the category of the product intended to be purchased, our second research question of interest will instead require product characteristics to be used to predict the corresponding product category instead. Additionally, a key finding is the possible reduction in prediction accuracy when excluding textual data such as product descriptions, in the logistic regression model.

Usage of KNN in E-Commerce Recommendation Systems

Recommendation systems are the core of all e-commerce systems today. These systems deploy algorithms which are designed to create a unique and personalized online shopping experience for all users. They work by filtering information from a variety of user-based data sources including purchase and browsing history. (Kong et al., 2024). In this study, a k-nearest neighbours (KNN) classification model is used to construct the recommendation system. Compared to other models, it has the shortest training time but with poor performance. This can be attributed to the model's inability to capture complex patterns between users and their purchases (Kong et al., 2024).

In the case of our study, there is no data which describes the relationship between users and products besides user reviews, which will not be considered for our analysis. Our analysis relies solely on using product characteristics to predict the product category and as a result, the limitation addressed by Kong does not apply.

Influence of Product Characteristics on Online Preferences

The evolution of e-commerce systems has created the need to understand the factors which have the most influence in shaping the online shopping preferences of consumers. Such research can help a business determine the viability of an e-commerce model. Frequent users of e-ecommerce platforms were asked to complete a survey which was designed to collect data on the various characteristics of online products and how these related to their desire to shop online. Lee uses a binary decision tree model to predict the preference to shop online as supposed to in-person. The purchase frequency and price showed to have the highest influence on the decision to shop online (Lee et al., 2007). In addition, the decision tree model offered higher performance accuracy compared to logistic regression.

A key difference between this study and ours is the use of a decision tree model for binary versus nonbinary classification. This study however, enforces the use of such a model with similar product input features to analyze e-commerce sales data.

Classifying Product Descriptions Using TF-IDF

Term frequency-inverse document frequency (TF-IDF) is a statistical weight measuring technique which is used to both measure and rank important words from a textual document (Dagobert et al., 2023). In this study, the output of such an algorithm is fed into a support vector machine (SVM) classification system to predict the product category given a product description. Feature extraction for the SVM is completed by constructing a TF-IDF matrix after applying tokenization and stopword removal preprocessing techniques to e-commerce product descriptions. The SVM reported high accuracy highlighting the effectiveness of the TF-DF algorithm in detecting relevant terms in product descriptions.

In contrast, our third research question focuses on using product descriptions to identify similar products. This study enforces the use of the TF-IDF statistical method to evaluate the importance of words in product descriptions, allowing for us to proceed with this method to identify similar products using an IR model.

Identifying Customer Trends Using EDA

E-commerce platforms generate millions of transactions daily which contain valuable information about the ways consumers interact with the different products available for purchase. Conducting an exploratory data analysis (EDA) can help to identify relationships between consumer behaviour and product sales (Chakma et al., 2025). The results of such an analysis can be useful for advanced statistical manipulation including collaborative filtering models for recommendation systems. Chakma observes statistical summaries of the data such as variable counts, means, and percentiles in addition to creating bar plots to visualize customer age distributions. Simple mathematical computations are also carried out, such as average purchase cost by age group.

In another study focused on understanding retail patterns for Amazon India, heat maps are constructed to visualize sales by factoring in variables like state, category and garment size. Time-based analysis is also done through line graphs to analyze sales trends by month and day. These high-level analytics revealed key insights related to consumer behaviour, such as Maharashtra being the state with both the highest revenue and consumer activity when compared to others (Ali et al., 2026).

While our first research question does not require the use of any temporal data, we will still be using product data to compare differences among the various product categories. These studies have demonstrated the effectiveness of utilizing EDA methods in the context of an e-commerce setting to highlight key information prior to further statistical analyses.

Methodology

Preliminary Analysis

Extract descriptive statistics such as means and frequencies. Compute correlations to observe potential relationships between variables. Visualize variable distributions to observe general trends and outliers using boxplots and histograms.

Classification Models

Based on the results of the preliminary analysis, construct three different product category classifiers: a linear multinomial logistic regression model, and two non-linear classifiers: a k-nearest neighbours model and a decision tree model. Compare the three models using evaluation metrics such as accuracy, precision and recall.

Information Retrieval Models

Construct multiple unigram and bigram vector space models to process product descriptions. Apply different text preprocessing techniques (stopword removal, lemmatization, etc.) and use both the TF and TD-IDF statistical weight measures to evaluate importance of words. Calculate accuracy, precision and recall to evaluate search relevance.

Data Profiling

The *Amazon Sales Dataset* will be the primary source of data for this project which contains over 1000 authentic Amazon product records scraped using Python. The dataset contains 16 features, and *Figure 1* provides an overview of the relevant characteristics of the dataset.

Figure 1: Overview of Features

Feature	Description	Type
product_name	String containing product name	Text
category	Pipe separated string containing product categories	Text
discounted_price	Discounted price of product in Indian Rupees (INR)	Numerical with units
actual_price	Price of product in INR	Numerical with units
discount_percentage	Percentage discount on product	Numerical with units
rating	Rating of product from 1 to 5	Ordinal, categorical
rating_count	Number of individuals who voted for a product rating	Numerical
about_product	Description of product	Text

Figure 2: Overview of Data

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories|Accessories&Peripherals|...	₹399	₹1,099	64%	4.2	24,269	High Compatibility : Compatible With iPhone 12...
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories|Accessories&Peripherals|...	₹199	₹349	43%	4.0	43,994	Compatible with all Type C enabled devices, be...
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	Computers&Accessories|Accessories&Peripherals|...	₹199	₹1,899	90%	3.9	7,928	【 Fast Charger& Data Sync】-With built-in safet...
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories|Accessories&Peripherals|...	₹329	₹699	53%	4.2	94,363	The boAt Deuce USB 300 2 in 1 cable is compati...
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	Computers&Accessories|Accessories&Peripherals|...	₹154	₹399	61%	4.2	16,905	[CHARGE & SYNC FUNCTION]- This cable comes wit...

Figure 3: Data Types and Missing Values

#	Column	Non-Null Count	Dtype
0	product_id	1465 non-null	object
1	product_name	1465 non-null	object
2	category	1465 non-null	object
3	discounted_price	1465 non-null	object
4	actual_price	1465 non-null	object
5	discount_percentage	1465 non-null	object
6	rating	1465 non-null	object
7	rating_count	1463 non-null	object
8	about_product	1465 non-null	object
9	user_id	1465 non-null	object
10	user_name	1465 non-null	object
11	review_id	1465 non-null	object
12	review_title	1465 non-null	object
13	review_content	1465 non-null	object
14	img_link	1465 non-null	object
15	product_link	1465 non-null	object

The dataset contains no missing values and the quantitative features will need to be converted to their numerical equivalent data types.

Data Cleaning

Characters and punctuation will be removed from the numerical features before converting them to their float equivalents.

Figure 4: Cleaned Dataset

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories|Accessories&Peripherals|...	399.0	1099.0	64.0	4.2	24269.0	High Compatibility : Compatible With iPhone 12...
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories|Accessories&Peripherals|...	199.0	349.0	43.0	4.0	43994.0	Compatible with all Type C enabled devices, be...
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	Computers&Accessories|Accessories&Peripherals|...	199.0	1899.0	90.0	3.9	7928.0	【 Fast Charger& Data Sync】-With built-in safet...
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories|Accessories&Peripherals|...	329.0	699.0	53.0	4.2	94363.0	The boAt Deuce USB 300 2 in 1 cable is compati...
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	Computers&Accessories|Accessories&Peripherals|...	154.0	399.0	61.0	4.2	16905.0	[CHARGE & SYNC FUNCTION]- This cable comes wit...

Summary of Features

Figure 5: Summary of Quantitative Features

	actual_price	discounted_price	discount_percentage	rating_count
count	1465.000000	1465.000000	1465.000000	1463.000000
mean	5444.990635	3125.310874	47.691468	18295.541353
std	10874.826864	6944.304394	21.635905	42753.864952
min	39.000000	39.000000	0.000000	2.000000
25%	800.000000	325.000000	32.000000	1186.000000
50%	1650.000000	799.000000	50.000000	5179.000000
75%	4295.000000	1999.000000	63.000000	17336.500000
max	139900.000000	77990.000000	94.000000	426973.000000

The price of products ranges from ₹39 to ₹139,900 with the average price being about ₹5445. This suggests that there are extreme outliers present in both the actual and discounted price features.

Figure 6: Overview of Product Categories

category	count
Computers&Accessories|Accessories&Peripherals|Cables&Accessories|Cables|USBCables	233
Electronics|WearableTechnology|SmartWatches	76
Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	68
Electronics|HomeTheater,TV&Video|Televisions|SmartTelevisions	63
Electronics|Headphones,Earbuds&Accessories|Headphones|In-Ear	52
...	...
Home&Kitchen|Kitchen&HomeAppliances|SmallKitchenAppliances|RotiMakers	1
Home&Kitchen|Heating,Cooling&AirQuality|Parts&Accessories|FanParts&Accessories	1
Home&Kitchen|Kitchen&HomeAppliances|SmallKitchenAppliances|StandMixers	1
Home&Kitchen|Heating,Cooling&AirQuality|Fans|PedestalFans	1
Home&Kitchen|Kitchen&HomeAppliances|Vacuum,Cleaning&Ironing|Vacuums&FloorCare|VacuumAccessories|VacuumBags|HandheldBags	1

211 rows × 1 columns

There are 211 product categories 205 of which correspond to only one product. To address this massive data imbalance, only the categories with 30 or more products will be used for the analysis.

Figure 7: Product Categories with 30 or More Products

category	count
Computers&Accessories|Accessories&Peripherals|Cables&Accessories|Cables|USBCables	227
Electronics|WearableTechnology|SmartWatches	70
Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	68
Electronics|HomeTheater,TV&Video|Televisions|SmartTelevisions	61
Electronics|Headphones,Earbuds&Accessories|Headphones|In-Ear	47
Electronics|HomeTheater,TV&Video|Accessories|RemoteControls	39

Figure 8: Overview of Product Ratings

	count
rating	
4.1	244
4.3	230
4.2	228
4.0	129
3.9	123
4.4	123
3.8	86
4.5	75
4	52
3.7	42
3.6	35
3.5	26
4.6	17
3.3	16

Although Figure 8 does not display all possible values of ratings, several correspond to fewer than 30 products. Similarly to product categories, only the ratings which correspond to greater than 30 categories will be considered as per the law of large numbers.

Figure 9: Product Ratings with 30 or More Products

	count
rating	
4.1	102
4.2	101
4.3	94
4.0	70
3.9	54

Note that the counts differ as these are the ratings with 30 or more products after filtering the dataset to only include categories with 30 or more products.

Measures of Central Tendency

Figure 10: Averages by Product Category

category	actual_price	discounted_price	discount_percentage	rating_count
Computers&Accessories|Accessories&Peripherals|Cables&Accessories|Cables|USBCables	824.469787	305.710745	60.579787	15163.984043
Electronics|Headphones,Earbuds&Accessories|Headphones|In-Ear	2427.437500	932.250000	53.062500	111749.125000
Electronics|HomeTheater,TV&Video|Accessories|RemoteControls	1126.833333	386.666667	57.277778	848.166667
Electronics|HomeTheater,TV&Video|Televisions|SmartTelevisions	38955.169492	24383.966102	37.627119	12739.000000
Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	20613.138462	15773.938462	23.092308	37995.200000
Electronics|WearableTechnology|SmartWatches	9181.762712	2429.423729	69.864407	26073.508475

Smart televisions have the highest average price compared to all other categories while smart watches have the highest offered average discount.

Figure 11: Averages by Product Rating

rating	actual_price	discounted_price	discount_percentage	rating_count
3.9	5900.542963	3417.106296	61.111111	15387.814815
4.0	4193.957143	2394.284143	54.628571	16884.257143
4.1	9970.901961	6498.057745	46.019608	54020.588235
4.2	11581.019802	6349.376238	54.495050	26996.485149
4.3	17382.691489	11120.735106	49.351064	10128.819149

The products with the highest rating also have the highest average price. In contrast, the products with the lowest rating have the highest average discount.

Figure 12: Median Rating by Product Category

category	rating
Computers&Accessories|Accessories&Peripherals|Cables&Accessories|Cables|USBCables	4.1
Electronics|Headphones,Earbuds&Accessories|Headphones|In-Ear	4.1
Electronics|HomeTheater,TV&Video|Accessories|RemoteControls	4.0
Electronics|HomeTheater,TV&Video|Televisions|SmartTelevisions	4.2
Electronics|Mobiles&Accessories|Smartphones&BasicMobiles|Smartphones	4.1
Electronics|WearableTechnology|SmartWatches	4.1

Smart televisions have the highest median rating while remote controls have the lowest median rating among all categories. It is important to note, however, that due to the compression of the dataset earlier, all selected product categories are reporting a median rating of 4 or higher.

Data Distributions

Figure 13: Histogram of Product Price

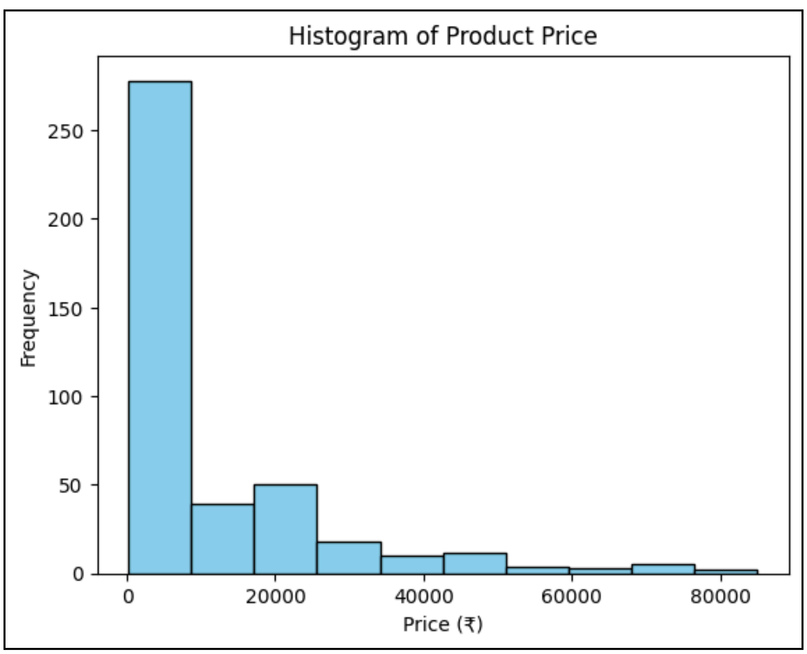
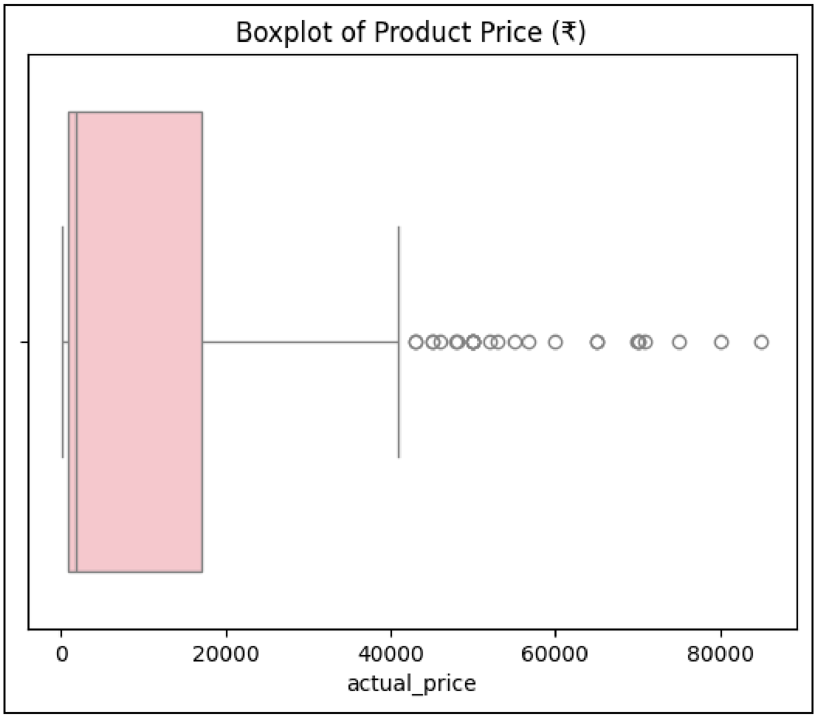


Figure 14: Boxplot of Product Price



The product price appears to be positively skewed due to many extreme data points.

Figure 15: Histogram of Discounted Product Price

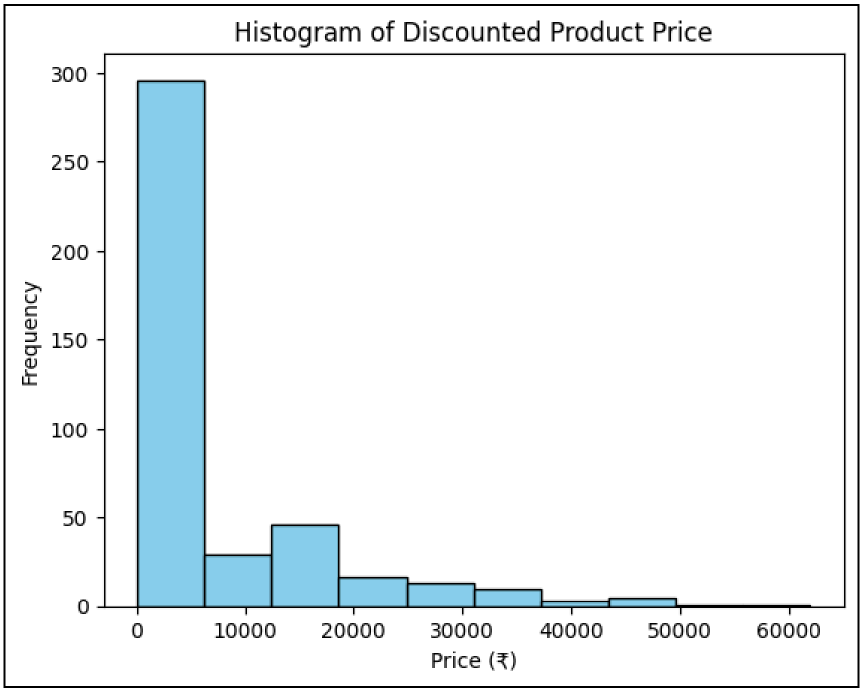
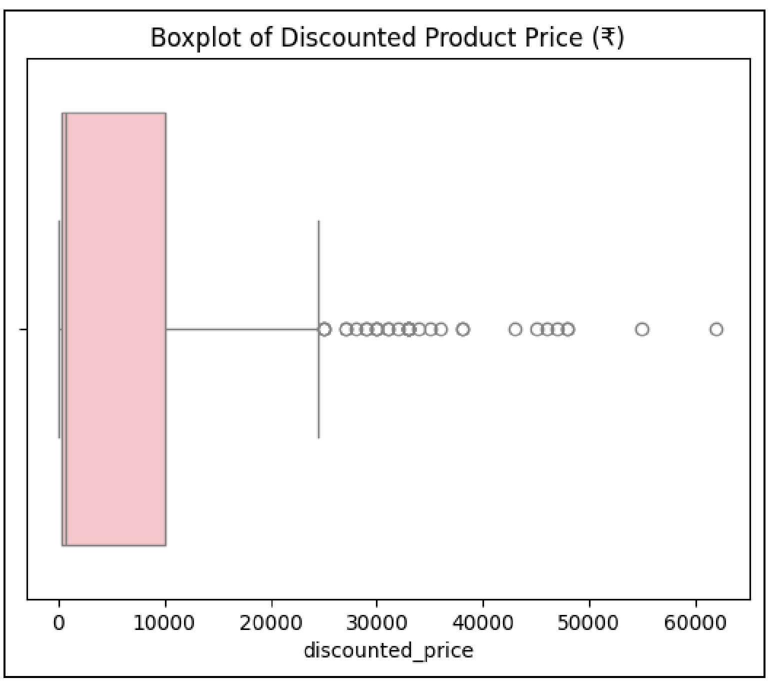


Figure 16: Boxplot of Discounted Product Price



Similarly to the distribution of product price, the discounted price is also positively skewed.

Figure 17: Histogram of Product Discount Percentage

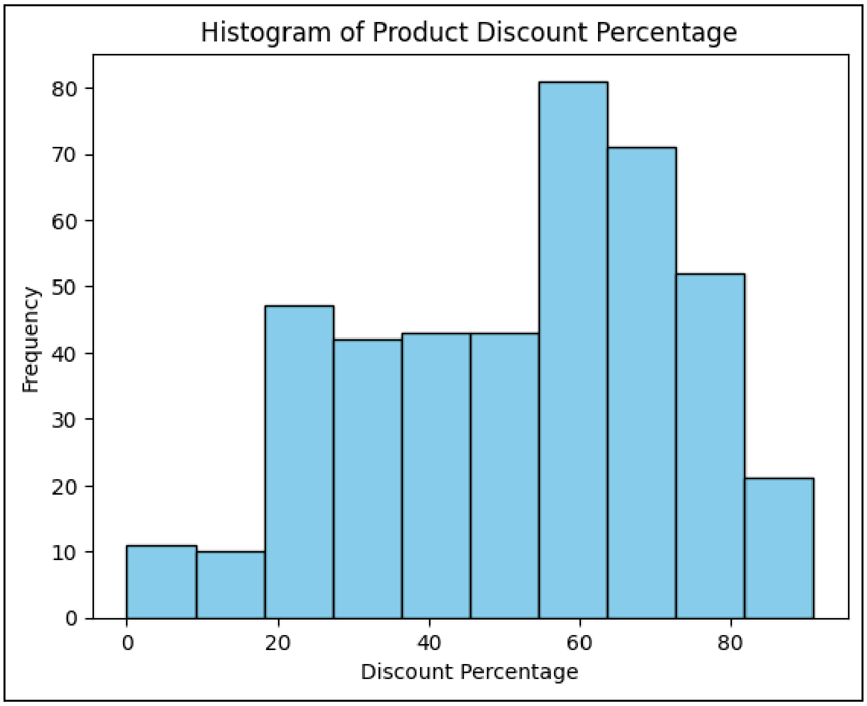
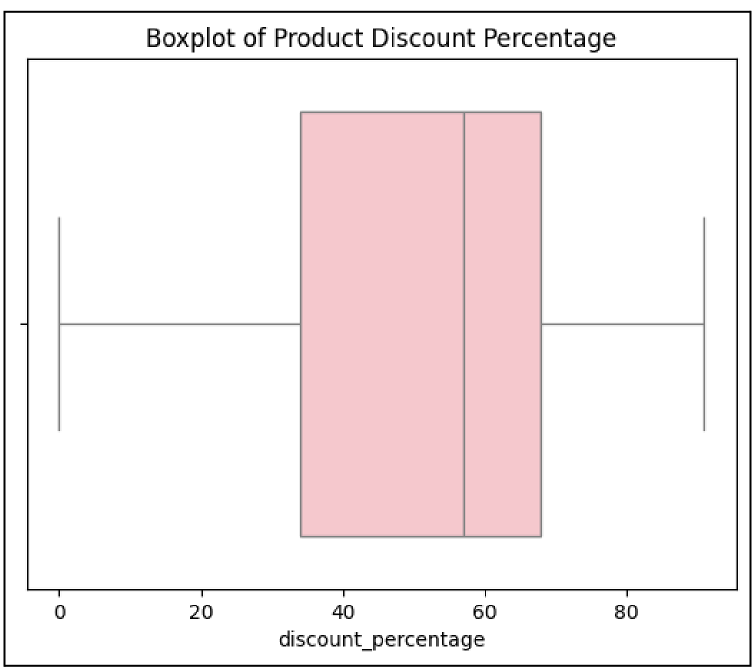


Figure 18: Boxplot of Product Discount Percentage



The distribution of the product discount percentage has a slight negative skew but is otherwise symmetric.

Figure 19: Histogram of Product Rating Votes

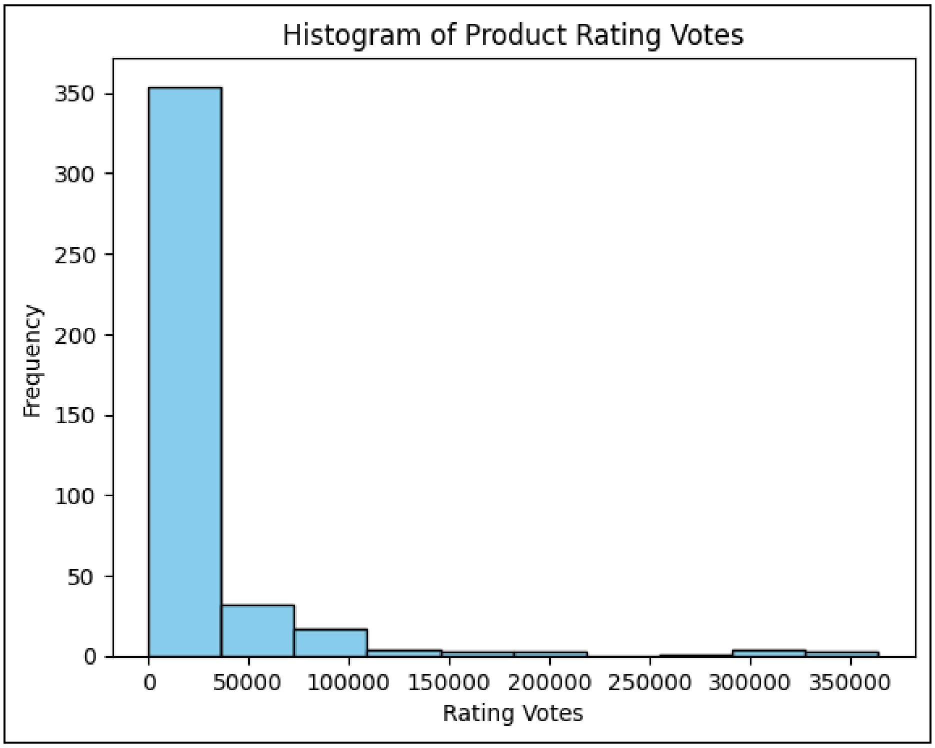
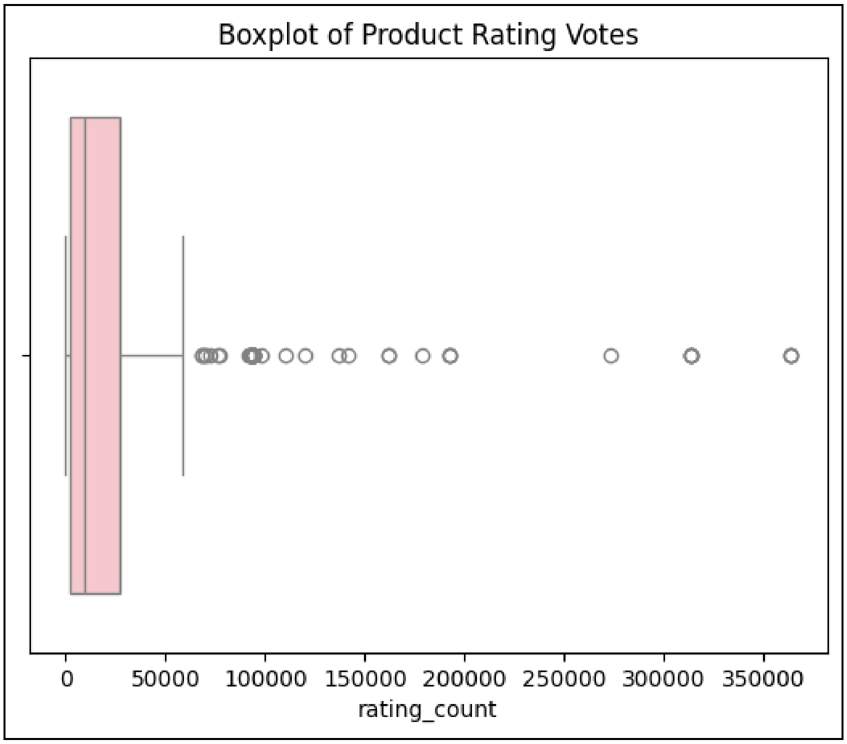
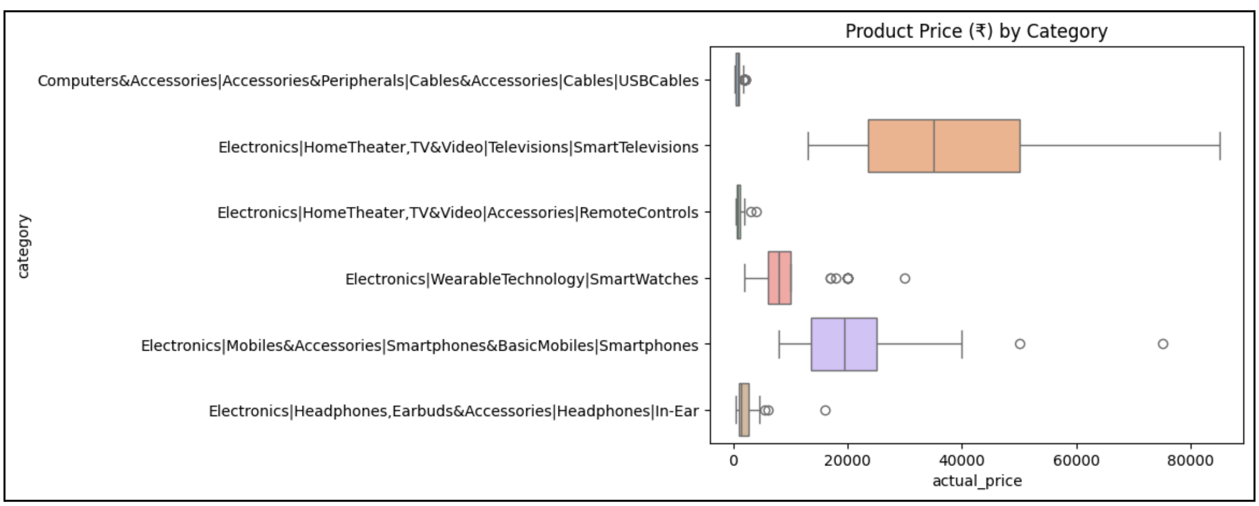


Figure 20: Boxplot of Product Rating Votes



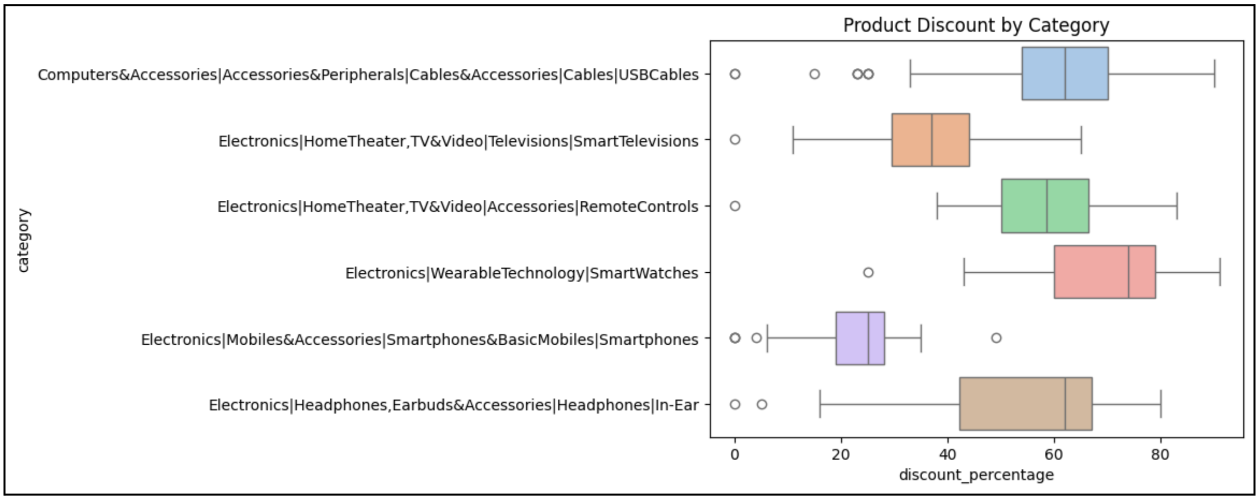
The distribution of the product rating votes has a strong positive skew, again indicating the presence of outliers.

Figure 21: Distribution of Product Price by Product Category



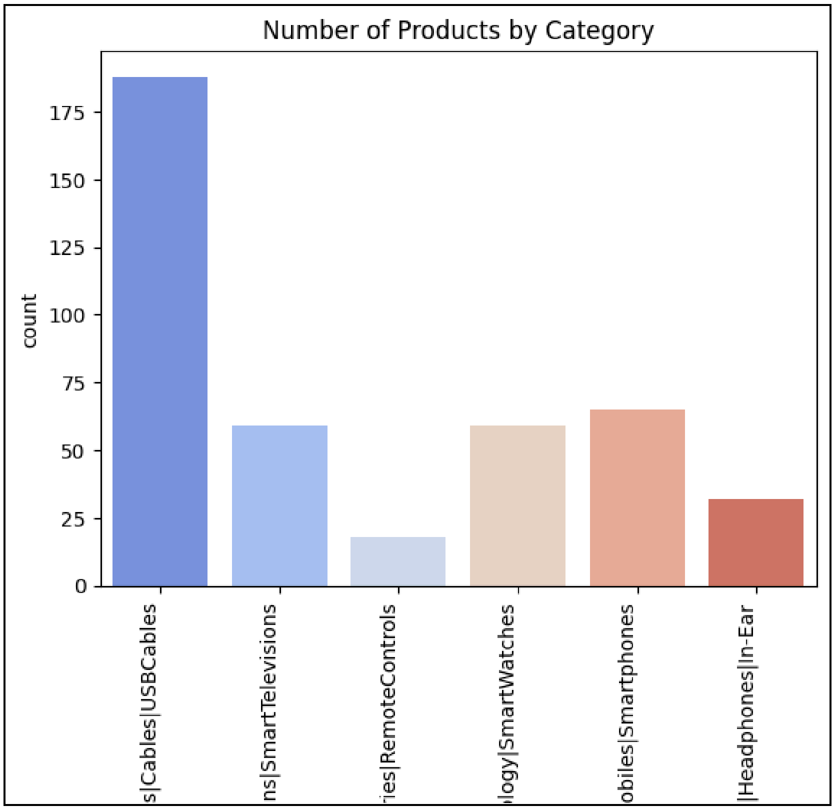
The prices of smart televisions and smartphones appear to be nearly symmetrically distributed with the fewest outliers among all categories. These two categories also have the highest product prices.

Figure 22: Distribution of Product Discount Percentage by Product Category



Smart watches have the highest offered discount while smartphones have the lowest.

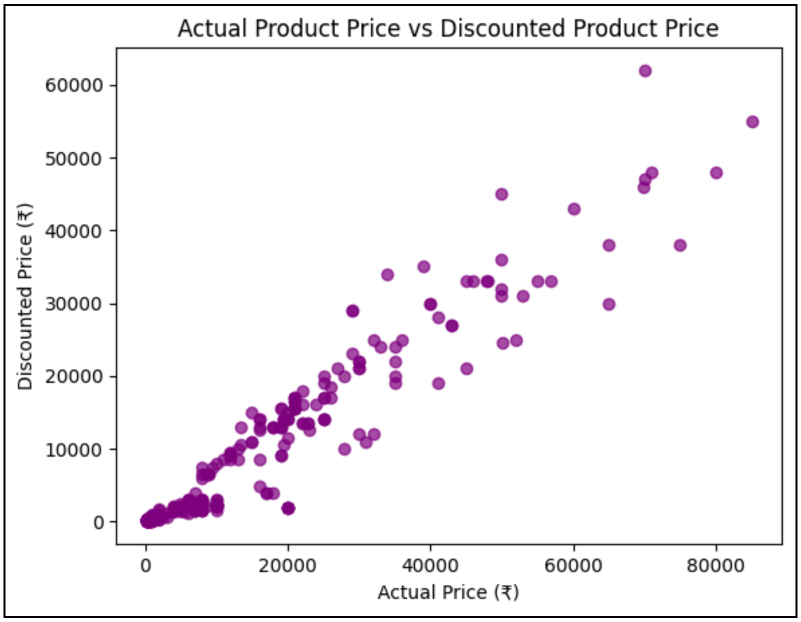
Figure 23: Distribution of Product Category



The majority of the products in the dataset are USB cables while there is very little data on remote controls.

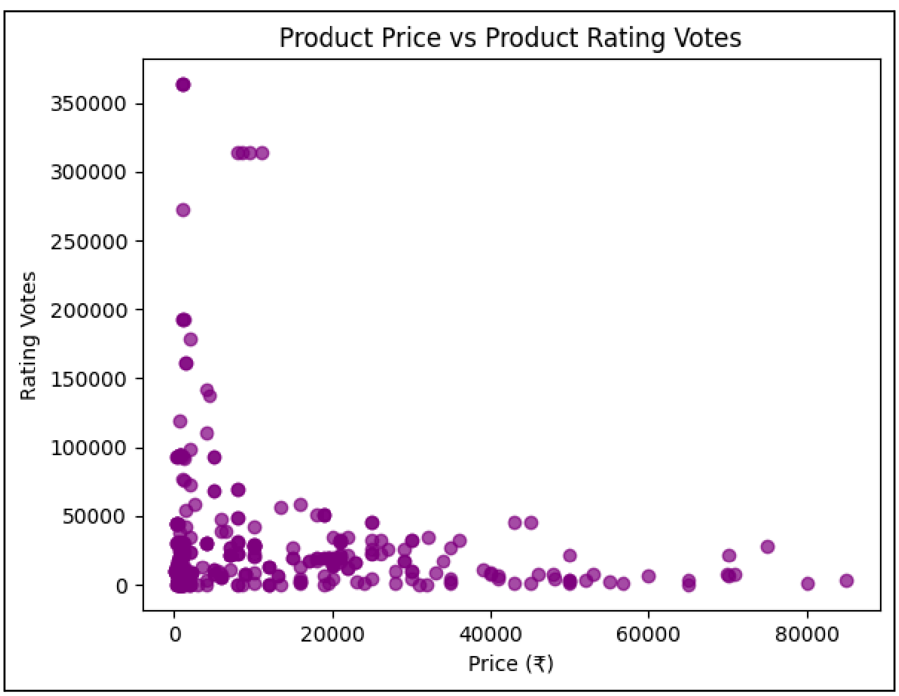
Detection of Linear Relationships

Figure 24: Relationship Between Actual Product Price and Discounted Product Price



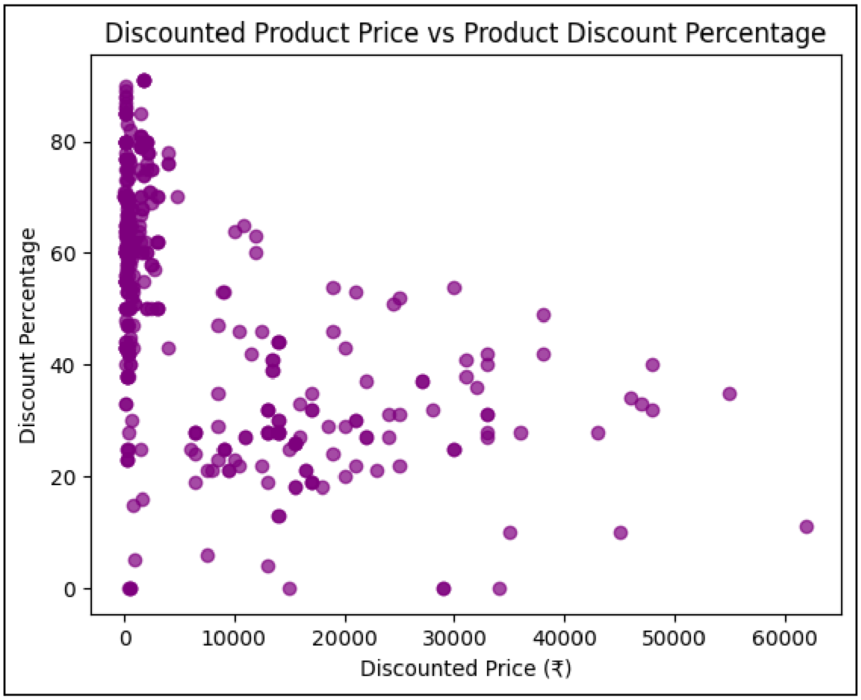
There appears to be a strong positive linear relationship between the actual and discounted product prices.

Figure 25: Relationship Between Product Price and Product Rating Votes



There appears to be a weak negative relationship between the product price and rating votes.

Figure 26: Relationship Between Discounted Product Price and Product Discount Percentage



The plot suggests a potential moderate negative linear relationship between the discounted product price and the discount being offered.

Figure 27: Correlations Between Quantitative Features

	actual_price	discounted_price	discount_percentage	rating_count
actual_price	1.000000	0.962910	-0.415841	-0.100234
discounted_price	0.962910	1.000000	-0.554204	-0.084120
discount_percentage	-0.415841	-0.554204	1.000000	-0.080742
rating_count	-0.100234	-0.084120	-0.080742	1.000000

Product price and discounted product price have a high correlation coefficient of about 0.96 indicating a strong potential linear relationship. A correlation of -0.42 suggests a moderate negative linear relationship between price and discount percentage while a correlation of -0.55 also implies the same between discounted price and discount percentage.

Data Structure and Preparation

The data for this project is structured as it is displayed in a tabular format and lacks any hierarchical structure. Not only is the dataset large, but it also contains a high level of class imbalance for the product category and product rating features. Both of these features have several data points which correspond to fewer than 30 products. For this reason and as previously discussed, the data has been filtered to include only those categories and ratings which relate to more than 30 unique products. As we are removing data points with very few observations, the resulting dataset is considered to be representative of the entire population of products.

Limitations

- Prices are in Indian Rupees which reduce the interpretability for the North American *Analytics* team.
- No information is provided regarding the purchase history for each product, making it difficult to identify which products require more attention from the marketing and sales departments.
- After filtering the product categories and ratings, the remaining data entirely corresponds to products with various subcategories in the electronics department with a rating of 3.9 or higher. This affects the ability to deliver insights for products with lower ratings and belonging to other categories.
- The product price features have a significant number of outliers, which will need to be accounted for prior to statistical modelling.
- Features which do not appear to have a normal distribution, such as price and rating counts may require further transformation to be suitable for a logistic regression model. Such transformations can include taking the logarithm or square root, which can reduce interoperability when examining the model results.

Privacy & Ethics

- The dataset has a Creative Commons (CC) license which provides the necessary permissions to freely use the data for non-commercial purposes.
- The dataset includes the names and user IDs of individuals who contributed to product reviews. Sensitive information such as this does not concern the scope of this project and therefore efforts will be made to ensure this data does not get shared, modified or reproduced in any way.

Architecture

Preprocessing

Thus far, the dataset has been prepared for the analysis by cleaning the numerical data of any unwanted characters and filtering the product category and rating data to include only records which are representative of the population. There is no need to handle missing values or invalid records as there are none present within the data. As for the outliers, the KNN and logistic regression classifiers would be the most sensitive, while decision trees are already robust.

Feature Engineering

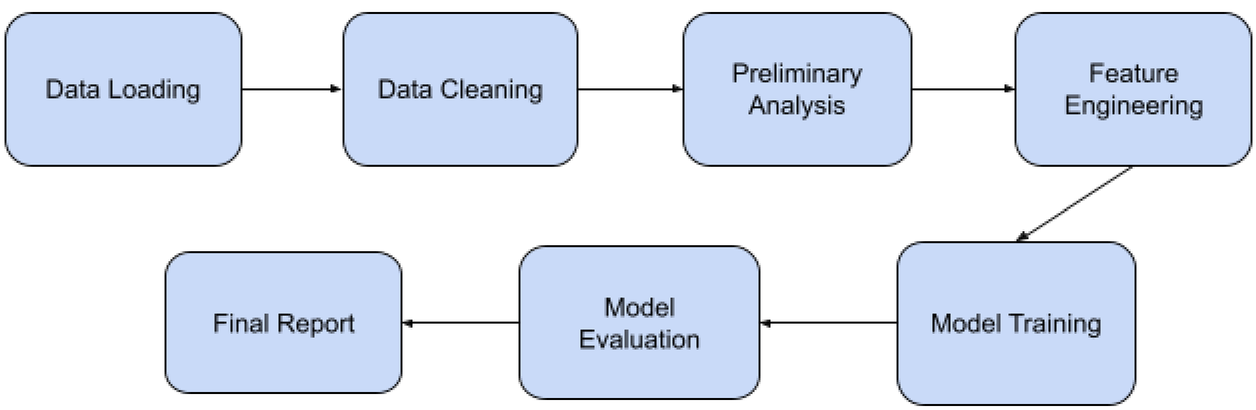
Building upon our previous discussion, due to the distance and weight based algorithms of KNN and logistic regression respectively, the treatment of outliers as missing values by applying imputation (TOMI) technique will be used. Instead of removing the outliers from the data, this method first requires the conversion of outliers to missing values, which are then imputed by the median for maximum machine learning efficiency (M. Seliem, 2022).

Furthermore, if there appears to be no linearity between the log odds of the target and predictors, an appropriate transformation such as taking the logarithm or square root will be applied to conform to the model assumptions of logistic regression. One-hot encoding will also be applied to the categorical features to preserve semantic meaning in the analysis.

Information retrieval models will require a different feature transformation approach. The product descriptions will undergo tokenization prior to the application of standard text preprocessing techniques. These steps will ensure the textual data is ready to be used as input in such a model.

Modelling and Evaluation

Figure 28: System Architecture Diagram



1. **Data Loading**: Download dataset from Kaggle and load into Python
2. **Data Cleaning**: Remove characters from numerical features and filter data to build a representative subpopulation
3. **Preliminary Analysis**: Obtain data summaries and generate relevant data visualizations
4. **Feature Engineering**: Apply feature transformation techniques to handle outliers and prepare categorical data

5. **Model Training**: Train the classification and information retrieval models on a subset of the data
6. **Model Evaluation**: Calculate performance metrics for model comparison
7. **Final Report**: Summarize visualizations, insights and recommendations into a consolidated document for stakeholders

Classification Models

After filtering the data to create a representative subpopulation of products, the remaining dataset retains only 30% of its original size. This shrink justifies the use of a smaller k-fold cross validation training strategy to avoid overfitting while capturing sufficient product data patterns. To prevent data leakage, any mathematical transformations will be computed using only the training data.

Three classifiers will be built: a linear multinomial logistic regression model and two nonlinear classifiers including KNN model and a decision tree. These models will take price, discounted price, discount percentage, rating and rating votes as predictors to predict the product category. Our second research question of interest requires the creation of a product classifier for the marketing team to select the most appropriate marketing content when promoting a new product based on its category.

The models will be compared using standard evaluation metrics such as accuracy, precision and recall. It is important to maintain balanced class data to maximize the interpretation of these metrics and to fairly compare model performance.

Information Retrieval Models

TF and TF-IDF matrices will be created using the product descriptions. A vector space information retrieval model will then be created by using these matrices and the description of a new product in order to output a ranked list of similar products already available for purchase. Such a model is required for the third research question to act as a tool for the sales teams to identify similar products for recommendation purposes.

Reproducibility

All work pertaining to this project can be accessed via the following GitHub repository: https://github.com/harman-khehara/Amazon-Product-Analysis.

GenAI Disclosure

The entirety of this project will be completed without the use of any generative artificial intelligence. Development history of the project will be tracked using GitHub and Google Suite. Justification of all work can be provided upon request.

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